CDS DS 210 Project	
Project Overview	Using the Iris dataset to predict the species of a flower given measurements like petal and sepal length. Dataset: https://www.kaggle.com/datasets/uciml/iris
Data Processing	Split the dataset into ⅔ train ⅓ test and random shuffled as data was sorted by species
Code Structure	Modules: split into (data, algorithm, helper funcs) - dataloader.rs: for loading Iris dataset and getting splits - math.rs: for any math functions not pre made - knn.rs: for the algorithm and its helper funcs Predictor Struct: Takes labels of nearest neighbors and gets most probable with .sample() KNN: Takes data and searches for the closest points with Euclidean distance. Then uses .sample() from the Predict Struct to get the most probable label. Workflow: Dataloader gets matrix and labels which are given to knn, then the test dataset is used as input to KNN.predict() which gets the labels of nearest neighbors and gives to Prediction.sample(). Prints test accuracy.
Test	Tests that the Predictor finds the most probable token with a dummy case Tests that the KNN finds the nearest neighbors with a dummy case Output: running 2 tests test tests::test_prediction ... ok test tests::test_knn ... ok test result: ok. 2 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.00s
Outputs	cargo run 32 (Just right) KNN is 96% accurate cargo run 100 (Over fit) KNN is 30% accurate cargo run 10 (under fit, this case is good as simple) KNN is 98% accurate

How to use	Use cargo run ___ ←number neighbors (default 32) Expected run time > 1 sec